Student ID_____ Student Name_____

IS221: Web Applications Development

School of Information Technology, Engineering, Mathematics and Physics

Final Examination
Semester 1 2022

Face-To-Face and Online Mode

Duration of Exam: 3 hours + 10 minutes

Reading Time: 10 minutes

Writing Time: 3 hours

INSTRUCTIONS

1. This examination has two sections:
   a. Section A: Coding Questions (40 marks)
   b. Section B: Short and Long Answer Questions (60 marks)
2. **All** questions are compulsory
3. Write **ALL** answers in the spaces provided within this paper.
4. This exam is **marked out of 100** and is worth **50%** of your **overall course marks**.
5. To **pass** this paper you require **at least 40% for this exam** which is equivalent to **40 marks out of 100**
6. This paper has 22 pages including this page.

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Section A: Coding Questions 40 Marks

Question 1: ASP.Net

Web forms let you build interactive web pages that collect information from a user and process it on the web server. The Web page in figure 1 below shows an example of a typical web form. Use figure 1 web form and later information to answer the questions that follow.

Figure 1: Web Form

a. Write the ASP code for txtCusName Textbox. **(2 marks)**

b. Write the ASP code for ddlLocation Dropdown List in asp.net with 3 List items (Ba, Lami, Lautoka). **(4 marks)**

c. Bedroom option has a dropdown list (ddlBedrooms). The following are the options in ddlBedrooms: One, Two, Three, Four, Five or more. This list uses Bedrooms.xml as a data source. Write the XML code for Bedrooms.xml. **(5 marks)**

d. Figure 2 below represent the RentHouse table in Microsoft Access. What control is used to connect the web form to the RentHouse table in Microsoft Access? **(1 mark)**

RentHouse

Field Name	Data Type
ID	AutoNumber
Cus_Name	Short Text
Cus_Phone	Short Text
Location	Short Text
Bedroom	Short Text
Pet_Friendly	Yes/No
Bills_Included	Yes/No

Figure 2: RentHouse Table

e. The customer phone number should only accept numeric values and have a total of seven digits.

i. What control should be used to validate txtCusPhone number? **(1 mark)**

ii. Write down the validation expression to make sure the customer does not input any invalid phone number. **(1 mark)**

iii. What control should be used to force the customer to enter a phone number before submitting the form? **(1 mark)**

f. Upon clicking the "Request for Information" (btnRequestInfo) button, the customer information will be inserted into the database. Referring to figure 3 write the missing VB code that will handle btnRequestInfo click event. Part of the codes of the Event Handler is given below. **(7 marks)**

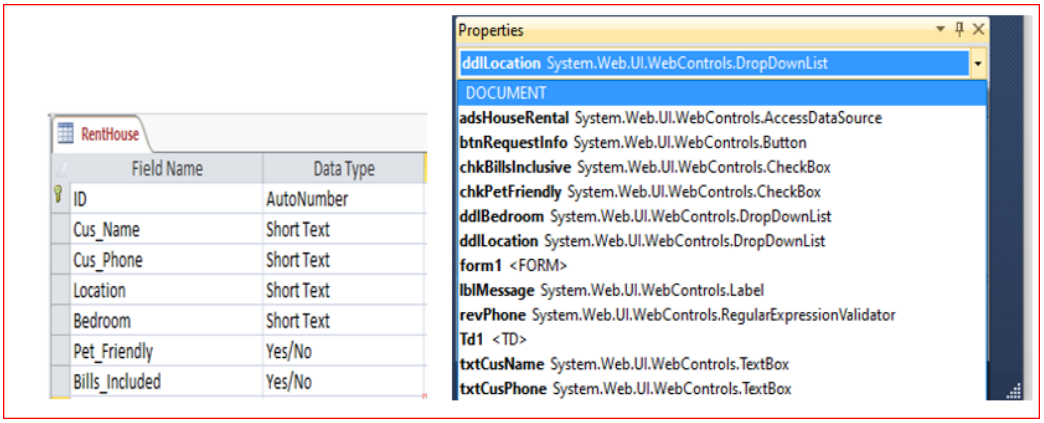

Figure 3

```
Protected Sub btnRequestInfo_Click(sender As Object, e As
System.EventArgs)Handles btnRequestInfo.Click

' write the missing VB codes below
```

```
lblMessage.Text = "Thank you for choosing us. <br/> We will check the
availability and get back to you soon."
End Sub
```

g. Write the ASP code for Masoori Bungalows to have a printer friendly website by referencing the print.css file that will render successfully when a webpage is printed or viewed in print preview mode. **(2 marks)**

```
<link href="print.css" type="text/css" rel="stylesheet" media="print" />
```

h. Write a CSS style to hide everything that is inside the nav element. **(1 mark)**

```
nav {
display: none;
}
```

i. Masoori Bungalows wants to make the website tablet friendly (780px and below). Write a media query that can be used to write CSS for tablets. **(2 marks)**

```
@media only screen and (max-width: 780px) {...}
```

Question 2: Data Tables

The table below in Figure 4 shows Pet Medicine Product Price List for KOKOtree pet medicine products & Services Company.

a. Write the appropriate HTML codes that will markup the table in Figure 4 below to make meaningful semantic table. [Hint: consider all those tags we've covered: the table, the row, the data, the headers, the table head, table body, table foot, caption, the col group and the col, row tags]. **(7 marks)**

KoKotree Pet Medicine Product Price List

			Retail		Case	
Item #	**Product or Service**	**Price (ea.)**	**Retail Price (Case)**	**Case Discount**	**12+ Case Price**	**12+ Case Discount**
100050	Advance Pet Oral Care Toothbrush and Toothpaste	$9.55	$108.87	$5.73	$103.14	$11.46
100043	Basic Teeth Cleaning and Exam	$100.00	$1,140.00	$60.00	$1,080.00	$120.00
100013	Calm Cat Anxiety Relief Spray	$9.49	$108.19	$5.69	$102.49	$11.39
	Totals of these columns	$119.04	$1357.06	$71.42	$1285.63	$142.85

Figure 4

b. Write style rules that can be applied to the table in figure 4 by adding the following properties:

- All borders collapse.
- Table border: solid 1px #3F2B21
- Table cell bottom border: solid 1px #CCAE51
- Table cells padding: 0.5rem
- Caption: bold, left aligned, color #CCAE51, font size 2rem and padding of 1rem.

(6 marks)

```
table {
border: 1px solid #3F2B21;
border-collapse: collapse;
}

th, td{
border: 1px solid #CCAE51;
padding: 0.5rem;
}

caption{
font-weight: bold;
text-align: left;
color: #CCAE51;
font-size: 2rem;
padding: 1rem;
}
```

Section C: Short/Long Answer Questions 60 Marks

Question 1

List three ways to control information overload when designing a web site. **(3 marks)**

-create manageable information segments
-control page length
-use hypertext to connect facts, concepts and relationships

Question 2

Explain why you would include both graphic and text-based links on a web page. **(2 marks)**

for easily understandable
easy download and browser cache
more appealing

Question 3

You have been hired to develop your local supermarket's website and are required to integrate the following site requirements:

- Logo and mission statement should be featured.
- Search box.
- Links to the other pages of the website should be available.
- Weekly specials should be a main feature content.
- Advertisements of related products, deals and news should also feature.
- Copyright statement should feature as well.
- Uniform look and feel for all the pages, the only difference should be the actual content.

a. With the aid of a diagram, illustrate how the home page template would look like.

(5 marks)

b. With the aid of a diagram, illustrate how you would design a responsive version of the homepage for mobile devices that have screen sizes ranging from 320px to 768px.

(5 marks)

Question 4

You have been asked to develop a website for a company that is proposing to sell flowers online. Using the web development lifecycle discuss how you would go about implementing this project. Your discussion should include the key stages of the cycle and examples of activities and artefacts involved in that stage. **(10 marks)**

Stage 1 requirement and specification (1/2 Mark)

Activities (1 Mark)

determines how the customer/user wants to use the website, for instance, will there be payments, credit cards, etc

Artefacts (1/2 Mark)

registered domain such as www.flowery.com

Stage 2 information design and taxonomy creation (1/2 Mark)

Activities (1 Mark)

sketching the web and it's pages and the navigation in relation to them, etc.

Artefacts (1/2 Mark)

sketching the webpages and how it should be interlinked with the content

Stage 3 graphics design and page template creation (1/2 Mark)

Activities (1 Mark)

creatin web graphics and page templates, for instance, logo in flower online shop which is a representative of the shop

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Artefacts (1/2 Mark)

web graphics and page templates

Stage 4 construction and web development (1/2 Mark)

Activities (1 Mark)

coding of the required elements and content
construction of database

Artefacts (1/2 Mark)

database driven functioning website with forms, components, and catalogues

Stage 5 publishing and promotion (1/2 Mark)

Activities (1 Mark)

determining whether the cost of the onsite or web host company
use facebook for website promotion

Artefacts (1/2 Mark)

fully functional website

Question 5

Suppose you are part of the web designing team for the university's website. How would you go about designing and developing USP's website so that it is easy to maintain and update; yet each school can customize the design to reflect their school colors and requirements. **(5 marks)**

generate the master pageuse external style sheets and link to mast

Question 6

ASP.NET applications processing is depicted in figure 5. Using the labels 1 – 5, describe this process. **(5 marks)**

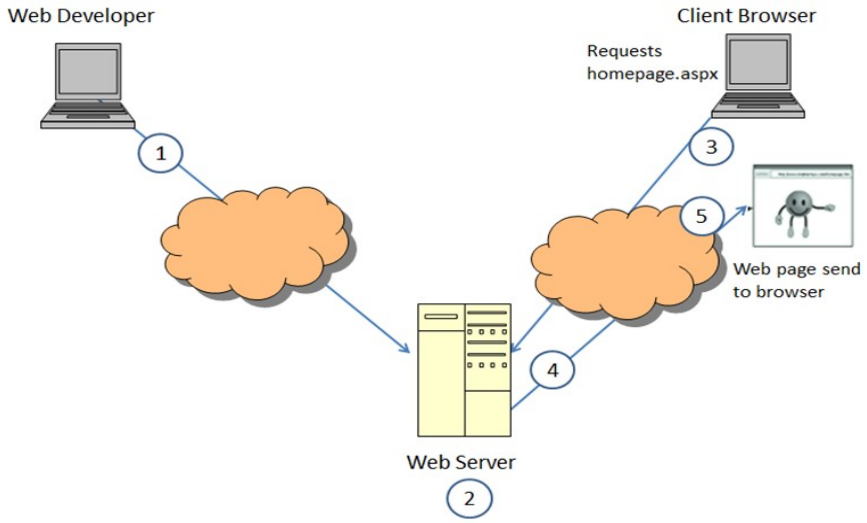

Figure 5: ASP.Net applications processing

(1) the developer creates the web page and it is stored in the web server(2) programming code can b

Question 7

Effective navigation should provide information that helps the user to answer the following questions:

- Where am I?
- Where can I go?
- How do I get there?
- How do I get back to where I started?

Compare and contrast how Facebook and YouTube implemented their navigation schemes and how they provided answers to the stated questions. For each question determine the required information to be provided and how each of the sites provided the answers. **(8 marks)**

Where am I?

Required information (1 Mark)
the page currently being viewed by the user

Facebook (1/2 Mark)
breadcrumb

YouTube (1/2 Mark)
breadcrumb

Where can I go?

Required information (1 Mark)
where the user is in relation to the rest of the website

Facebook (1/2 Mark)
related links

YouTube (1/2 Mark)
related links

How do I get there?

Required information (1 Mark)
easy to understand links

Facebook (1/2 Mark)
contextual linking

YouTube (1/2 Mark)
site map

How do I get back to where I started?

Required information (1 Mark)
alternatives to the browser's back button that lets the users return back to the orginal page

Facebook (1/2 Mark)
page turners

YouTube (1/2 Mark)
linked navigation

Question 8

Using good coding practices ensures the greatest standards-compliance, presentation, and usefulness of the content. List with examples five rules that conforms to good coding practices. **(5marks)**

all opening tags must have a end tagempty tags must be closedall elements t

Question 9

Using a diagram explain CSS box model. **(5 marks)**

Question 10

Discuss how you would design for both Accessibility and Legibility **(2 Marks)**

accessibility
-information and user interface to be easily understood
-user interface to be operable
-information to not be overloaded

responsive
-use breakpoints
-use media queries
-test the site on multiple browsers

Question 11

Moodle content management learning system is very compelling to use, attractive, easy to navigate and a very popular content management website. In designing the Moodle website, the designers took into consideration the following pointers:

- Crafting the look and feel structure, color, contrast, labelled sections
- Design portability use of different stylesheets, coding standards
- Easy access to your information according to the 3 clicks
- Planning for Clear presentation of information grid structure, active whitespace, clear heading
- Design for low bandwidth minimized use of graphics, reusing of universal links

For each consideration discuss how Moodle content management website implemented them.

(5 Marks)

Draft Page

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Draft Page

THE END